Software Processes

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Topics covered

1. What is a Software Process?
2. Software Process Activities
3. Waterfall Development
4. Iterative and Incremental Development
5. Other Process Models
    1. Spiral Development
    2. V Model

What is a Software Process

- Ivar Jacobson, Grady Booch, and James Rumbaugh
  - "A process defines who is doing what, when, and how to reach a certain goal"

- It is widely accepted that the quality of the product depends on the quality of the process!
  - Good processes lead to good software
  - Good processes reduce risk
  - Good processes enhance visibility

- Synonyms include software life cycle and software development process
  - There are several models for such processes, each describing approaches to a variety of tasks or activities that take place during the process

Variety of Software Processes

- Software products are very varied...
  - Therefore, there is no standard process for all software engineering projects
- BUT successful software development projects all need to address similar issues
  - This creates a number of process steps that must be part of all software projects

Basic Process Steps

1. Feasibility study
2. Requirements engineering
3. Design
4. Implementation
5. Verification and validation (acceptance and release)
6. Operation, maintenance, and evolution

❒ It is essential to distinguish among these aspects and to be clear which you are doing at any given moment

Feasibility Study

- A feasibility study precedes the decision to begin a project
  - What are the projected benefits?
  - What is the scope of the proposed project?
  - Is the project technically feasible?
  - What are the costs, timetable?

- A feasibility study leads to a decision
  - Go or no-go

Requirements Analysis and Definition

- The requirements analysis and definition establish the system's services, constraints and goals by consultation with users
  - They are then defined in a manner that is understandable by both users and development staff

- Requirements engineering process
  - Requirements elicitation and analysis
  - Requirements specification
  - Requirements validation

Design and Implementation

- The process of converting the requirements specification into an executable system

- Design and implementation
  - Design a software structure that realizes the specification
  - Translate this structure into an executable program

- The activities of design and implementation are closely related and may be inter-leaved

Verification and Validation

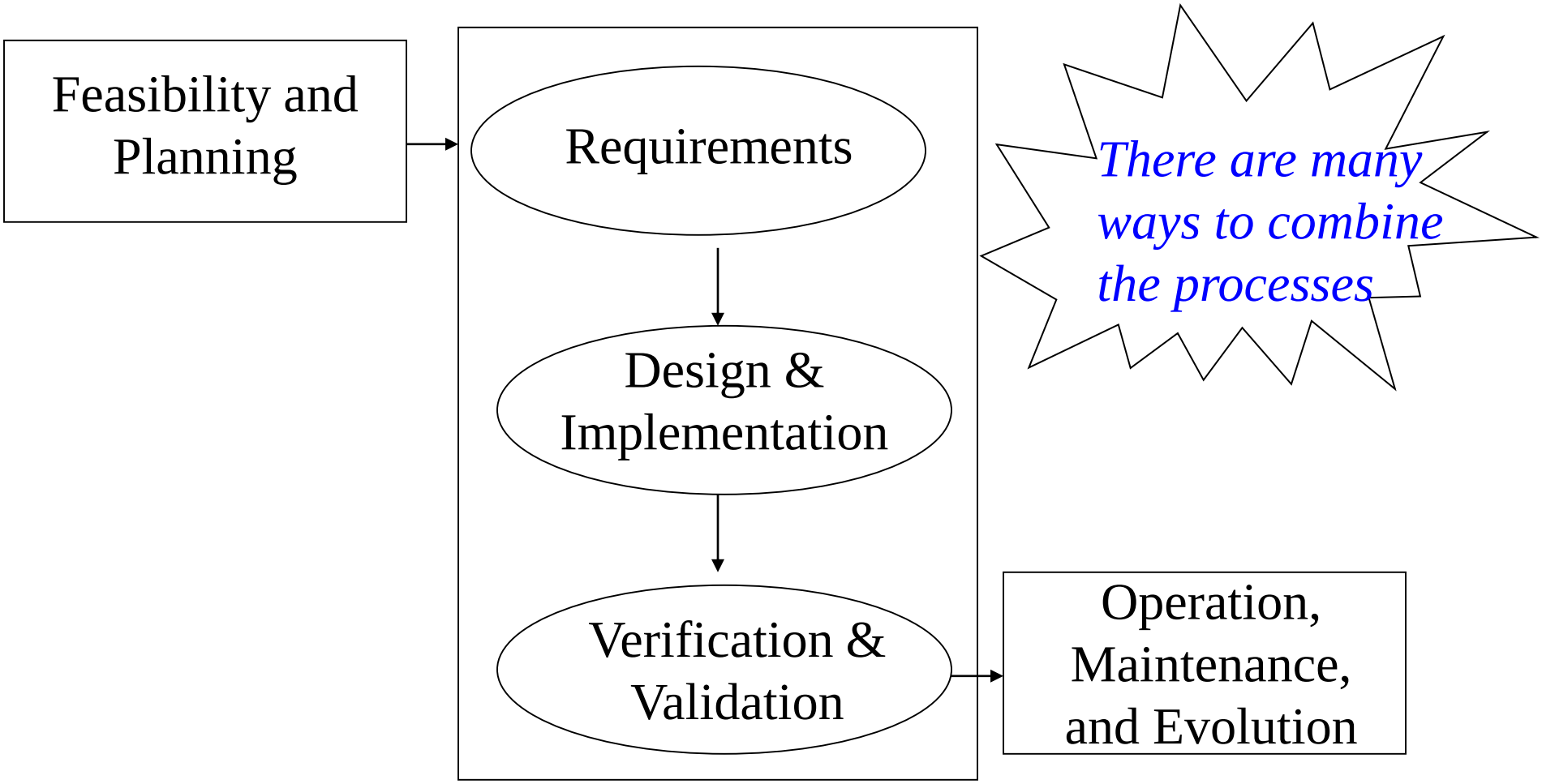
- **Verification and validation (V & V) is intended to show that a system conforms to its specification and meets the requirements of the customer**

- **Verification & validation**
  - **Involves checking and review, and testing**
  - **Testing involves executing the system with test cases that are derived from the specification of the real data to be processed by the system**
    - The complete system should be tested against the requirements

Operation, Maintenance, and Evolution

- Operation: The system is put into practical use
- Maintenance: Errors and problems are identified and fixed
- Evolution: The system evolves over time as requirements change, to add new functions or adapt the technical environment
- Phase out: The system is withdrawn from service

Combining the Process Steps

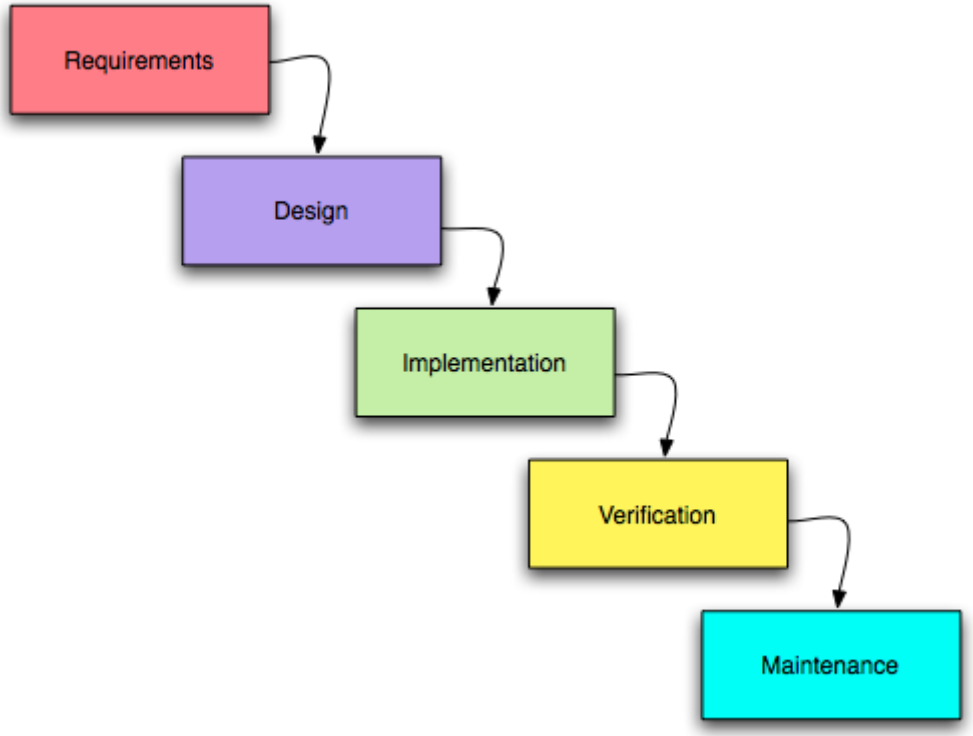


The Waterfall Model

- The waterfall model is a sequential software development model
  - Development is seen as flowing steadily downwards (like a waterfall) through the phases of requirements analysis, design, implementation, testing (validation), integration, and maintenance

- The origin of the term "waterfall" is often cited to be an article published in 1970 by Winston W. Royce (1929–1995) although Royce did not use the term "waterfall" in this article
  - Ironically, Royce was actually presenting this model as an example of a flawed, non-working model (Royce 1970)

The Waterfall Model



Discussions of the Waterfall Model

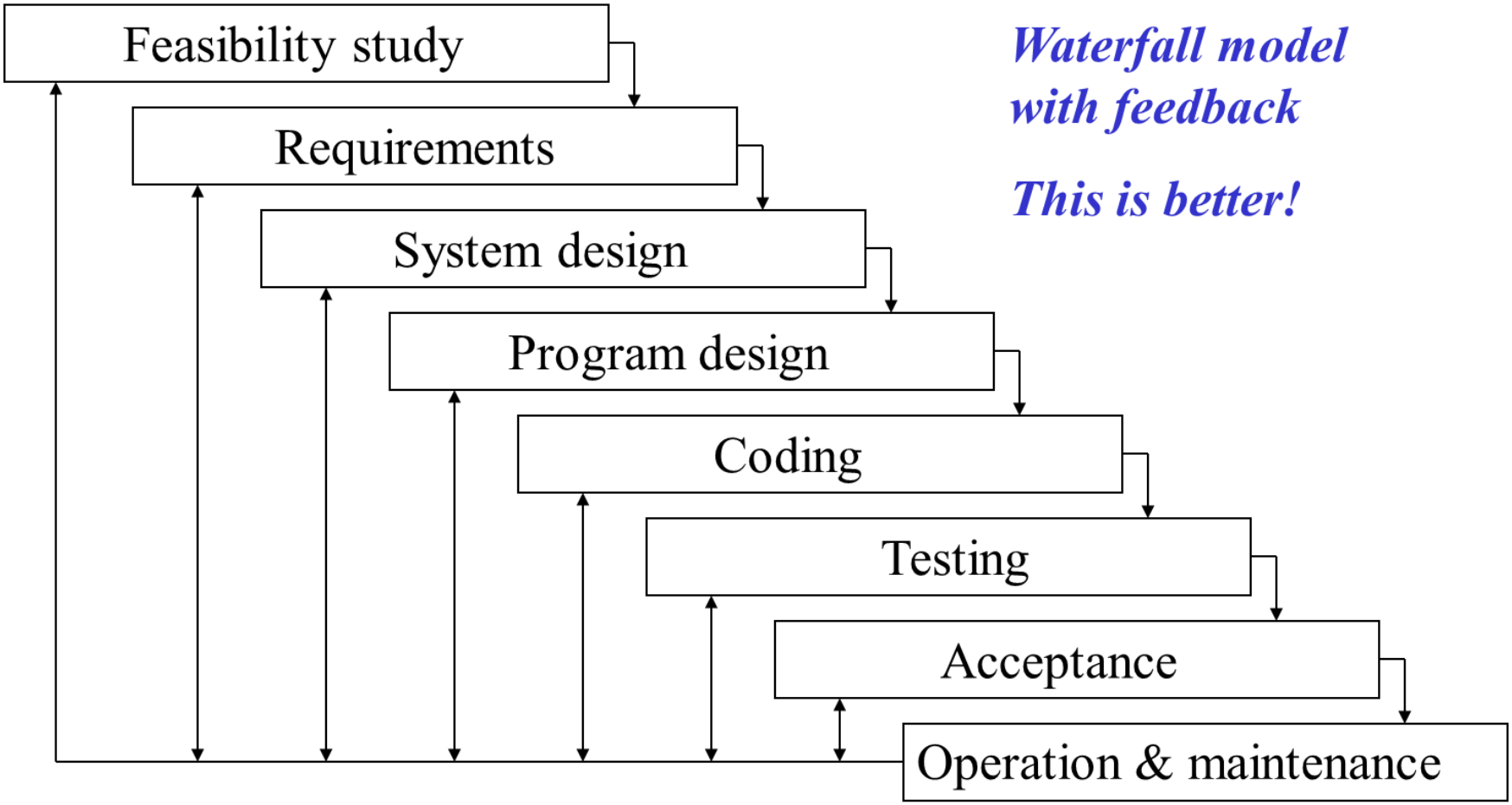
❒ Advantages:

- **This model is simple and easy to understand and use**
- **It is easy to manage due to the rigidity of the model**
  - Each phase has specific deliverables and a review process
  - Process visibility is high

❒ Disadvantages:

- **It is very difficult to go back and change something in earlier phases**
- **Not suitable for the projects where requirements are at a moderate to high risk of changing**
- **No working software is produced until late during the life cycle**

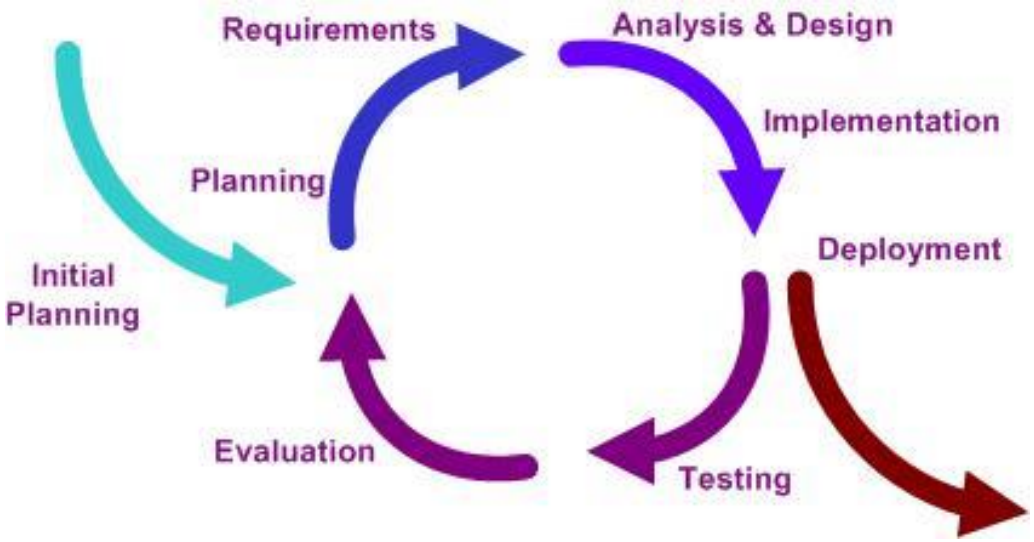
Modified Waterfall Model



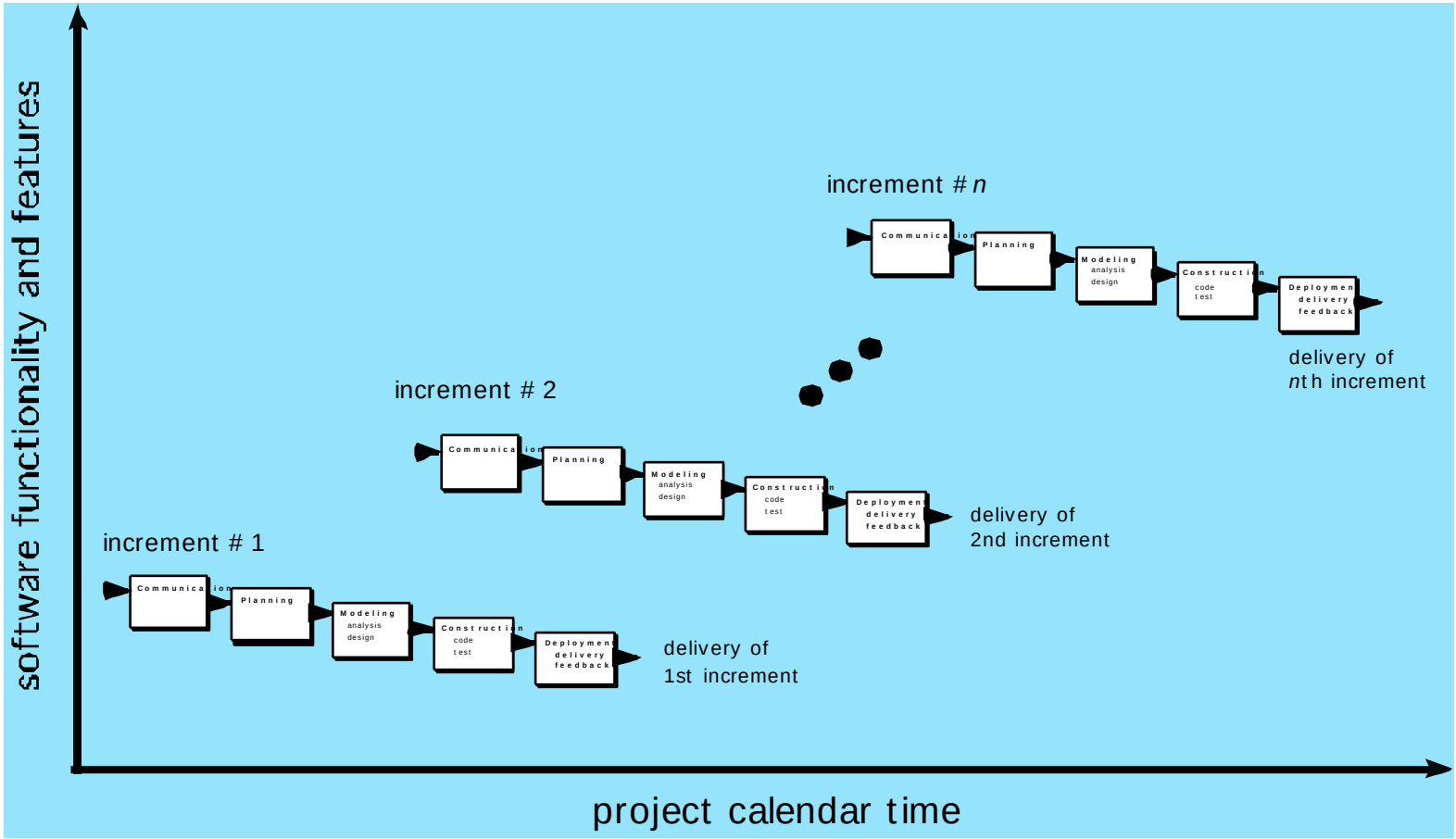
Waterfall model with feedback

This is better!

Iterative and Incremental Development



Iterative and Incremental Development



“Incremental” and “Iterative”

- Incremental development is a scheduling and staging strategy in which the various parts of the system, i.e., increments, are developed at different times or rates, and integrated as they are completed
  - The alternative to incremental development is to develop the entire system with a "big bang" integration

- Iterative development is a rework scheduling strategy in which time is set aside to revise and improve parts of the system
  - It does not presuppose incremental development, but works very well with it

Advantages and Disadvantages

❒ Advantages

- **Customer value can be delivered with each increment so system functionality is available earlier**
- **Early increments act as a prototype to help elicit requirements for later increments**
- **Lower risk of overall project failure**
- **The highest priority system services tend to receive the most testing**

❒ Disadvantages

- **Poor resource planning**
  - Difficult to predict efforts like cost, time and resources required at the beginning of a project
- **Poor software architecture and design**
  - Costly system architecture or design issues may arise because not all requirements are gathered up front for the entire lifecycle

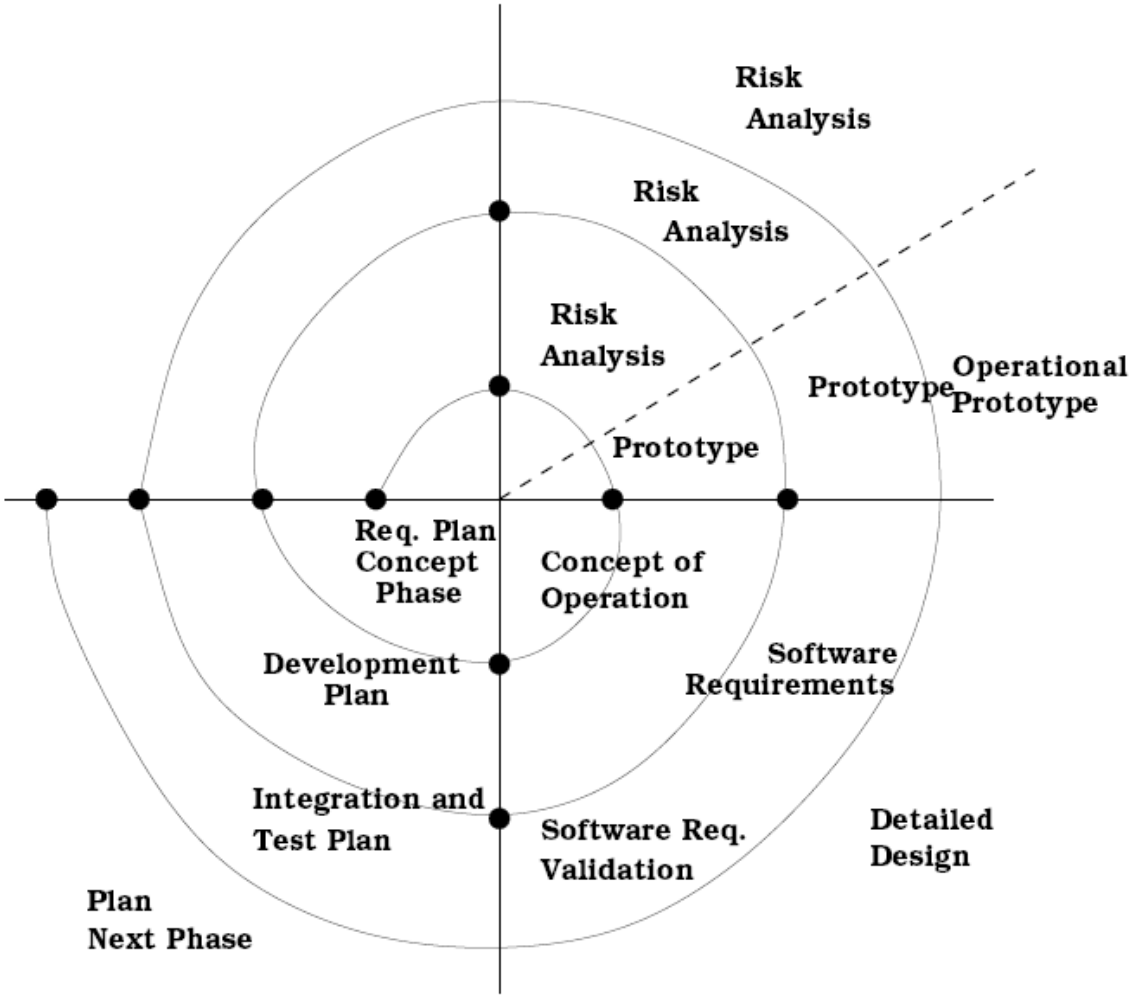
Other Software Processes

- Spiral Development
- V Model

Spiral Development

- Proposed by Barry Bohem, 1988
  - Process is represented as a spiral rather than as a sequence of activities with backtracking.
  - Each loop in the spiral represents a phase in the process
  - No fixed phases such as specification or design - loops in the spiral are chosen depending on what is required
  - Risks are explicitly assessed and resolved throughout the process

Spiral Development

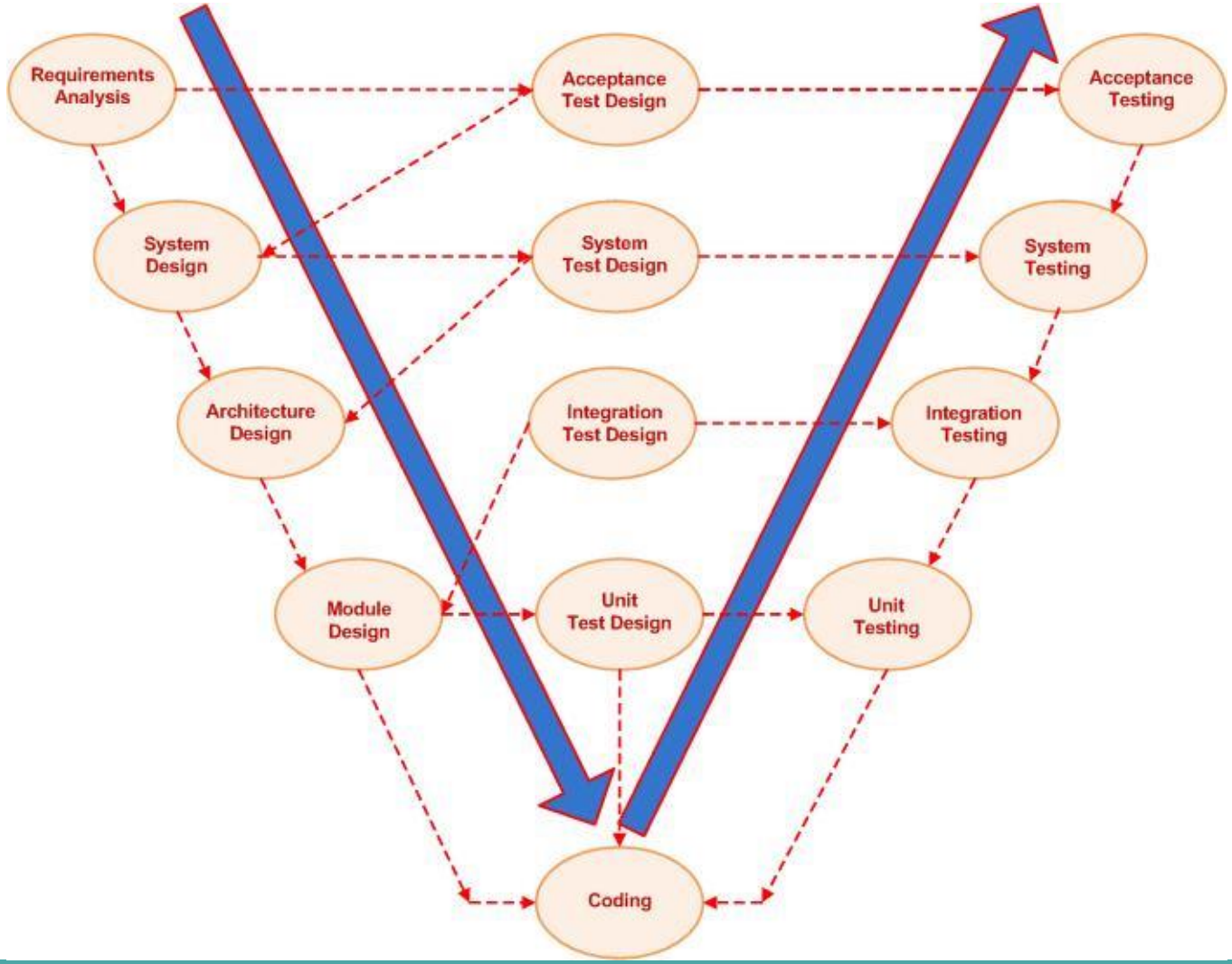


V Model

- The V-model is a software development model which can be presumed to be the extension of the waterfall model
  - The V-Model demonstrates the relationships between each phase of the development life cycle and its associated phase of testing

- The V-model can be said to have developed as a result of the evolution of software testing
  - The tests in the ascending (Validation) hand are derived directly from their design or requirements counterparts in the descending (Verification) hand
  - The 'V' can also stand for the terms Verification and Validation

V Model


Requirements Analysis
Acceptance Test Design
Acceptance Testing
System Design
System Test Design
System Testing
Architecture Design
Integration Test Design
Integration Testing
Module Design
Unit Test Design
Unit Testing
Coding


Q & A

개발 프로세스에 대한 고찰

- 기업에서 소프트웨어를 주먹구구식으로 개발할 수는 없다
  - 소프트웨어 개발에 필요한 일련의 절차와 방법이 존재해야 한다
  - 개발 프로세스가 없으면 관리와 통제가 불가능하다
- 소프트웨어 개발 프로세스는 단순하고 효율적이어야 한다
  - 과도하게 복잡해지면 비용이 증가하고 저항이 생긴다
- 소프트웨어 개발 프로세스가 개발 문화를 이길 수는 없다
  - 그러나 개발 문화를 천천히 바꿀 수는 있다
- 개발 단계별로 문서 산출물이 요구되는데, 많은 개발자들은 문서화 작업을 시간 낭비로 본다
  - 요구사항 문서는 계약서와 같다
  - 설계 문서는 가장 중요한 커뮤니케이션 수단이다
  - 구현물에 대한 문서화가 안되면 아무도 사용하지 않는다
    - 개발자 본인도 사용하기 어려워진다